SESSION CHECK · MARKER'S KEY

Session 1 — Where things are: the celestial sphere

Practical Astronomy Intensive

9 questions **15** minutes **8** concepts **9** diagnostic

Nine questions at the end of the first indoor session. They arrived able to define the celestial sphere and unable to point at the meridian; this checks whether the definitions have turned into something they could act on outside tonight. Angular measure is asked twice — everything later leans on it and nobody arrives with it.

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1 You hold a clenched fist out at arm's length against the sky. Roughly what angle does it span, and how does the full Moon compare?

`angular-measure` `Recall` `Multiple choice`

a About 45° — and the full Moon is roughly a coin's width at arm's length
REVEALS The full Moon as a coin at arm's length; grossly oversized hand-spans

b About 1° — and the full Moon is a few times larger than that
REVEALS No calibrated hand-spans; degrees not attached to anything on the sky

c About 10° — and the full Moon is far smaller, about half a degree, covered by a little fingernail at arm's length ✓

d About 10° — and the full Moon is about the same, roughly a fist's width
REVEALS The full Moon as a fist-sized object on the sky

> Fist ≈ 10°, thumb-width ≈ 2°, little fingernail ≈ 1°, full Moon ≈ 0.5°. The Moon feels huge and is not; every student should measure it tonight rather than believe it.

2 Mizar and Alcor sit about 12 arcminutes apart in the handle of the Big Dipper. A friend says 'so they must be practically touching out there.' What is wrong with that?

`angular-measure` `Reason` `Multiple choice`

a It is wrong only because 12 arcminutes is actually a large angle, so they are far apart in space
REVEALS Angular separation read as real separation, with the arithmetic also mis-scaled

b Angular separation is a difference in direction only — it says nothing about how far apart the two stars actually are in space ✓

c The Sun and Moon both span half a degree, so angles are the same for every object and mean nothing
REVEALS Angular size dismissed as meaningless rather than understood as direction

d Nothing is wrong — 12 arcminutes is a tiny angle, so they must be close together in space
REVEALS Angular separation read as real separation — direction confused with distance

> An angle on the sky is a difference of direction and nothing more. The Sun and Moon both span half a degree while the Sun is 400× wider — and 400× further. Two stars a few arcminutes apart may be hundreds of light-years apart along the line of sight.

3 A long-exposure photograph shows the stars drawn out into concentric arcs around a single point. What has the photograph actually recorded?

`diurnal-motion` `Apply` `Multiple choice`

- **a** The camera, bolted to the Earth, turning eastward through the exposure — the arcs are the Earth's rotation drawn out ✓
- b The stars orbiting the pole star, which sits at the centre of their motion
  REVEALS Polaris as a physical centre the stars orbit, rather than an accident of alignment with Earth's axis
- c The stars themselves sailing westward across the sky during the night
  REVEALS Stars as the moving objects; Earth's rotation not internalised
- d Every star in the frame rising and setting, since everything visible rises and sets
  REVEALS Everything rises and sets — circumpolar motion not recognised

> The sky is effectively fixed; the Earth turns eastward beneath it, so the whole sky appears to wheel westward around the celestial pole. Star trails are a picture of the photographer's own rotation.

4 What is the celestial sphere, and where do its poles and equator come from?

`celestial-sphere` `Apply` `Multiple choice`

- **a** A bookkeeping model of infinite radius that turns directions into geometry — its poles and equator are Earth's own axis and equator projected outward ✓
- b A real shell at a great distance with the stars fixed on it, all at the same distance from us
  REVEALS Celestial sphere taken literally — stars as objects on one physical shell
- c Another name for the sky dome — the visible half of the sky above your horizon
  REVEALS Celestial sphere collapsed into the local horizon dome; the projection idea missing
- d A model sphere, but with its poles and equator being genuine features of the sky, independent of Earth
  REVEALS Celestial poles/equator as properties of the sky rather than Earth's axis and equator projected out

> The sphere is not a claim about where stars are; it is a device for treating directions as points on a sphere so they can be computed. Its poles and equator are borrowed from Earth and thrown onto the stars.

5 A student writes in her log: 'Jupiter — altitude 34°, azimuth 118°.' What must she add for that entry to be usable by anyone, including her?

`altaz-coords` `Apply` `Multiple choice`

- a Nothing — altitude and azimuth are a fixed address for an object, like RA and Dec
  REVEALS Alt-az treated as a fixed address; the local, time-dependent nature of the system missed
- b Only the date — azimuth is fixed for an object and only the altitude drifts
  REVEALS Azimuth believed fixed; only altitude thought to change with time
- c The direction she was facing when she measured, since azimuth runs from there
  REVEALS Azimuth measured from the observer's facing rather than clockwise from due north
- **d** The date, time and her location — alt-az changes minute by minute and is meaningless without all three ✓

> Alt-az is a local, instantaneous pointing: it depends on where you stand and what time it is. Azimuth runs clockwise from due north — a sloppy north biases every pointing by the same angle. RA/Dec is the fixed address; alt-az is not.

6 You are standing in Bhopal (latitude 23° N) and want to show a group Polaris. Which statement is true, and is the most useful thing to tell them?

`pole-star` `Apply` `Multiple choice`

a It is a middling star, about 48th brightest — find it with the Pointers of Ursa Major, only about 23° above the northern horizon ✓

b It is the brightest star in the sky — just look north and pick out the brightest thing there
REVEALS Polaris as the brightest star in the sky

c It is the star nearest to Earth, which is why it barely moves in the sky
REVEALS Polaris's stillness attributed to proximity rather than alignment with the rotation axis

d It sits almost directly overhead, so look up near the zenith
REVEALS Pole star believed to hang overhead — altitude = latitude not held

> Two things kill more pole-star hunts than anything else: expecting a blazing star (Sirius outshines Polaris by miles — hence the Pointers) and expecting it overhead. Its altitude equals your latitude: 23° from Bhopal, on the horizon from the equator. It barely moves because it lies near Earth's rotation axis, not because it is close.

7 A student in Bhopal (23° N) finds that Ursa Major dips below his northern horizon in the evening. A friend in Srinagar (34° N) insists it never sets for her. Who is right?

`circumpolar-stars` `Reason` `Multiple choice`

a Both — which stars are circumpolar depends only on your latitude, and Srinagar's higher pole keeps more of Ursa Major above the horizon ✓

b The Bhopal student — every star rises and sets, so nothing can truly be circumpolar
REVEALS Everything rises and sets — circumpolar stars not accepted at all

c The Srinagar student — Ursa Major is circumpolar, and that is a fixed property of the constellation for everybody
REVEALS Circumpolarity treated as a property of the star, the same for all observers

d Neither — circumpolarity depends on the season, and they simply observed at different times of year
REVEALS Circumpolarity attributed to season rather than latitude

> A star is circumpolar if it lies within your latitude's angle of the celestial pole. From the equator no star is circumpolar; from the pole every visible star is. It is a fact about where you stand, not about the star.

8 Saturn is up all night. Why is it worth waiting for it to transit your meridian rather than observing it at 9 pm when it is low in the east?

`meridian-transit` `Apply` `Multiple choice`

a At transit it is at its highest, so you look through the least air — low in the east the same planet is a shimmering mush ✓

b Because the meridian is a line printed on the star chart, and objects are only in the catalogue's position when they cross it

REVEALS Meridian as a line on a chart rather than the observer's own north–south line

c There is no real difference — an object is equally worth observing at any time it happens to be above the horizon

REVEALS Altitude treated as irrelevant to image quality; atmospheric path length ignored

d Because at transit the planet crosses the celestial equator, which is where the atmosphere is thinnest

REVEALS Meridian muddled with the celestial equator

> The meridian is your own north–south line through your zenith — it moves when you do, and it is not printed on any chart. An object is highest as it transits, so it is seen through the shortest path of atmosphere. Everything else is a compromise.

9 An astronomer in London writes to a student in Chennai asking her to describe Canopus, the second-brightest star in the sky, since he has never seen it. Why has he never seen it?

`latitude-sky` `Reason` `Multiple choice`

a Canopus is only visible from the southern hemisphere, and Chennai is in the southern hemisphere

REVEALS Visibility treated as a hemisphere switch rather than a continuous function of latitude

b London is too far north for stars to rise and set at all — everything there just circles the pole

REVEALS Mid-latitude star paths confused with polar ones; slant vs vertical vs circling not distinguished

c He must simply have been unlucky with weather or timing — everyone on Earth sees the same stars eventually

REVEALS Belief that all observers see the same sky given enough time

d Canopus is far enough south that it never rises for London's latitude, while from Chennai it is a routine sight ✓

> Your latitude decides which sky you own. Canopus (declination about −53°) never clears London's horizon and never will; Chennai (13° N) sees it every winter. Star paths also change shape with latitude: vertical at the equator, slanted at mid-latitudes, circling at the poles.